THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 8	$(m+n)^2 - m^2$
2	**(a)** 6	Find b by Pythagoras
3	**(d)** $\frac{12}{5}$	$\frac{1}{2}ab = \frac{1}{2}ch$
4	**(b)** 14	Sides in ratio $a : b$
5	**(c)** $\frac{9}{5}$	$AB^2 = AD \cdot AC$
6	**(d)** 25	Side $= s$
7	**(b)** 21	$(m+n)^2 - m^2$
8	**(b)** 30	Find b by Pythagoras
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{2}{3} = \frac{4}{EC} \Rightarrow EC = 6$; R is the Basic Proportionality Theorem.
10	**(c)** A is true but R is false.	$AC^2 = 36 + 64 = 100$, so $AC = 10$: A true; the theorem holds for every right-angled triangle, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{3}{7}$ (1); $DE = 6$ cm (1).	1+1+1
12	Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{27}{75} = \frac{3^2}{x^2}$ (1); $x = 5$ cm (1).	1+1+1
13	Hypotenuse 5 cm (1); $\frac{1}{2}ab = \frac{1}{2}ch$ so $h = \frac{12}{5}$ cm (1).	1+1
14	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{2}{7}$ (1); $DE = 6$ cm (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 2:3$ (1); area ratio $= 4:9$ (1).	1+1+1
16	Altitude bisects the base: $h^2 = (6)^2 - 3^2 = 27$ (1+1); $h = 3\sqrt{3}$ cm (1).	1+1+1
17	$AC = 5$ cm (1); $AD = \frac{AB^2}{AC} = \frac{9}{5}$ cm, $DC = \frac{BC^2}{AC} = \frac{16}{5}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{12}{5}$ cm (1).	1+1+1
	OR: $\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 3:5$ (1); area ratio $= 9:25$ (1).	1+1+1
18	Altitude bisects the base: $h^2 = (8)^2 - 4^2 = 48$ (1+1); $h = 4\sqrt{3}$ cm (1).	1+1+1
	OR: $AC = 13$ cm (1); $AD = \frac{AB^2}{AC} = \frac{25}{13}$ cm, $DC = \frac{BC^2}{AC} = \frac{144}{13}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{60}{13}$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) AA (each has a right angle and the same sun-angle) (1). (ii) $\frac{h}{24} = \frac{1.5}{2} \Rightarrow h = 18$ m (1). (iii) $\frac{6}{s} = \frac{1.5}{2} \Rightarrow s = 8$ m (1). (iv) $\sqrt{18^2 + 24^2} = \sqrt{900} = 30$ m by the Pythagoras theorem (2).	1+1+1+2
20	Right triangle with legs 50 and 120 (1); distance $= \sqrt{2500 + 14400} = 130$ km (2); half-distance triangle is similar with ratio $1:2$, so the distances are in ratio $2:1$ (2).	1+2+2
	OR: Draw a horizontal line from the top of the shorter pole: right triangle with legs 27 and 36 (2); $x^2 = 729 + 1296$ (1); $x = 45$ m (2).	2+1+2

Mistakes to avoid:

- Writing the BPT as $\frac{AD}{AB} = \frac{AE}{EC}$; mix of a part-to-whole and a part-to-part ratio.
- Equating the *areas* ratio to the *sides* ratio instead of its square.
- Matching vertices in the wrong order while writing similarity ($\triangle ABC \sim \triangle PQR$ must have A to P, B to Q, C to R).
- Applying the Pythagoras theorem with the wrong side as hypotenuse (hypotenuse is opposite the right angle and is the longest side).

- Using the converse of Pythagoras without squaring all three sides, or comparing with the wrong side.
- Saying two triangles are similar because they have equal ratio of only two pairs of sides (SAS needs the included angle).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{25}{13}$	$AB^2 = AD \cdot AC$
2	**(c)** 49	Side $= s$
3	**(a)** 40	$(m+n)^2 - m^2$
4	**(b)** 60	Find b by Pythagoras
5	**(b)** $\frac{120}{17}$	$\frac{1}{2}ab = \frac{1}{2}ch$
6	**(c)** 22	Sides in ratio $a : b$
7	**(a)** $\frac{64}{17}$	$AB^2 = AD \cdot AC$
8	**(c)** 81	Side $= s$
9	**(d)** A is false but R is true.	$5^2 + 6^2 = 61 \neq 49 = 7^2$, so it is not right-angled: A false; R is the (true) converse of the Pythagoras theorem.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	The sides are in the ratio $1 : 2$ (SSS similarity), so A is true; R is true but does not explain the similarity.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ratio of areas = (ratio of sides)^2 (1); $\frac{32}{162} = \frac{4^2}{x^2}$ (1); $x = 9$ cm (1).	1+1+1
12	Hypotenuse 13 cm (1); $\frac{1}{2}ab = \frac{1}{2}ch$ so $h = \frac{60}{13}$ cm (1).	1+1
13	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{4}{9}$ (1); $DE = 8$ cm (1).	1+1+1
14	Ratio of areas = (ratio of sides)^2 (1); $\frac{20}{245} = \frac{2^2}{x^2}$ (1); $x = 7$ cm (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 1 : 4$ (1); area ratio $= 1 : 16$ (1).	1+1+1
16	Altitude bisects the base: $h^2 = (10)^2 - 5^2 = 75$ (1+1); $h = 5\sqrt{3}$ cm (1).	1+1+1
17	$AC = 17$ cm (1); $AD = \frac{AB^2}{AC} = \frac{64}{17}$ cm, $DC = \frac{BC^2}{AC} = \frac{225}{17}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{120}{17}$ cm (1).	1+1+1
	OR: $\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 2 : 7$ (1); area ratio $= 4 : 49$ (1).	1+1+1
18	Altitude bisects the base: $h^2 = (12)^2 - 6^2 = 108$ (1+1); $h = 6\sqrt{3}$ cm (1).	1+1+1
	OR: $AC = 25$ cm (1); $AD = \frac{AB^2}{AC} = \frac{49}{25}$ cm, $DC = \frac{BC^2}{AC} = \frac{576}{25}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{168}{25}$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Basic Proportionality (Thales) theorem (1). (ii) $\frac{6}{4} = \frac{9}{EC} \Rightarrow EC = 6$ m (1). (iii) $AB = 10$ m, $AC = 15$ m (1). (iv) $\triangle ADE \sim \triangle ABC$: $\frac{DE}{20} = \frac{6}{10} \Rightarrow DE = 12$ m (1); area ratio $= \left(\frac{6}{10}\right)^2 = \frac{9}{25}$ (1).	1+1+1+2
20	Right triangle with legs 90 and 120 (1); distance $= \sqrt{8100 + 14400} = 150$ km (2); half-distance triangle is similar with ratio $1 : 2$, so the distances are in ratio $2 : 1$ (2).	1+2+2
	OR: Distance walked $= 12$ m; let shadow $= x$ m (1); similar triangles: $\frac{x}{x+12} = \frac{3}{9}$ (2); $x = 6$ m (2).	1+2+2

Mistakes to avoid:

- Writing the BPT as $\frac{AD}{AB} = \frac{AE}{EC}$; mix of a part-to-whole and a part-to-part ratio.
- Equating the *areas* ratio to the *sides* ratio instead of its square.
- Matching vertices in the wrong order while writing similarity ($\triangle ABC \sim \triangle PQR$ must have A to P, B to Q, C to R).
- Applying the Pythagoras theorem with the wrong side as hypotenuse (hypotenuse is opposite the right angle and is the longest side).

- Using the converse of Pythagoras without squaring all three sides, or comparing with the wrong side.
- Saying two triangles are similar because they have equal ratio of only two pairs of sides (SAS needs the included angle).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{168}{25}$	$\frac{1}{2}ab = \frac{1}{2}ch$
2	**(d)** 16	Sides in ratio $a : b$
3	**(c)** $\frac{18}{5}$	$AB^2 = AD \cdot AC$
4	**(a)** 121	Side $= s$
5	**(a)** 45	$(m+n)^2 - m^2$
6	**(a)** 210	Find b by Pythagoras
7	**(c)** $\frac{420}{29}$	$\frac{1}{2}ab = \frac{1}{2}ch$
8	**(d)** 20	Sides in ratio $a : b$
9	**(a)** Both A and R are true and R is the correct explanation of A.	All angles of an equilateral triangle are $60°$, so two angles match; this is the AA criterion in R.
10	**(c)** A is true but R is false.	The ratio of areas equals the *square* of the ratio of sides, so $\sqrt{16/81} = 4/9$: A is true; R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hypotenuse 17 cm (1); $\frac{1}{2}ab = \frac{1}{2}ch$ so $h = \frac{120}{17}$ cm (1).	1+1
12	$\triangle ADE \sim \triangle ABC$ (1); $\dfrac{DE}{BC} = \dfrac{AD}{AB} = \dfrac{5}{8}$ (1); $DE = 20$ cm (1).	1+1+1
13	Ratio of areas $=$ (ratio of sides)^2 (1); $\dfrac{50}{72} = \dfrac{5^2}{x^2}$ (1); $x = 6$ cm (1).	1+1+1
14	Hypotenuse 25 cm (1); $\frac{1}{2}ab = \frac{1}{2}ch$ so $h = \frac{168}{25}$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\triangle AOB \sim \triangle COD$ (AA) (1); $\dfrac{AO}{OC} = \dfrac{AB}{DC} = 3:4$ (1); area ratio $= 9:16$ (1).	1+1+1
16	Altitude bisects the base: $h^2 = (14)^2 - 7^2 = 147$ (1+1); $h = 7\sqrt{3}$ cm (1).	1+1+1
17	$AC = 29$ cm (1); $AD = \frac{AB^2}{AC} = \frac{400}{29}$ cm, $DC = \frac{BC^2}{AC} = \frac{441}{29}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{420}{29}$ cm (1).	1+1+1
	OR: $\triangle AOB \sim \triangle COD$ (AA) (1); $\dfrac{AO}{OC} = \dfrac{AB}{DC} = 4:9$ (1); area ratio $= 16:81$ (1).	1+1+1
18	Altitude bisects the base: $h^2 = (18)^2 - 9^2 = 243$ (1+1); $h = 9\sqrt{3}$ cm (1).	1+1+1
	OR: $AC = 41$ cm (1); $AD = \frac{AB^2}{AC} = \frac{81}{41}$ cm, $DC = \frac{BC^2}{AC} = \frac{1600}{41}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{360}{41}$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The ladder (1). (ii) $\sqrt{17^2 - 15^2} = \sqrt{64} = 8$ m (1). (iii) $8 + 15 + 17 = 40$ m (1). (iv) New distance $= \sqrt{17^2 - 8^2} = \sqrt{225} = 15$ m (1); the foot moved $15 - 8 = 7$ m (1).	1+1+1+2
20	Draw a horizontal line from the top of the shorter pole: right triangle with legs 9 and 12 (2); $x^2 = 81 + 144$ (1); $x = 15$ m (2).	2+1+2
	OR: Distance walked $= 12$ m; let shadow $= x$ m (1); similar triangles: $\dfrac{x}{x+12} = \dfrac{2}{8}$ (2); $x = 4$ m (2).	1+2+2

Mistakes to avoid:

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